

COMPUTING FUNDAMENTALS: HARDWARE, SOFTWARE, NETWORKS AND CLOUD

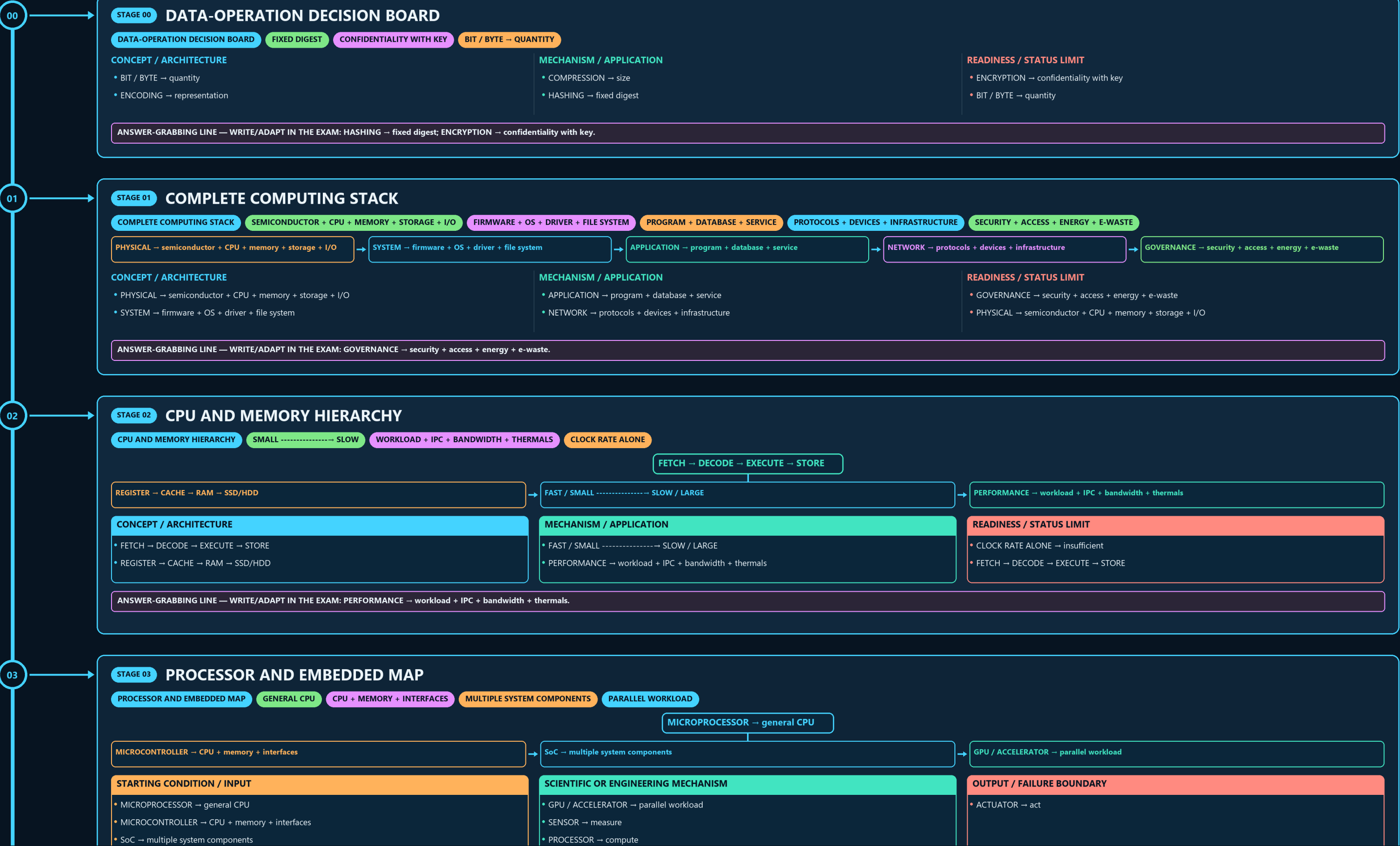
DATA-OPERATION DECISION BOARD → COMPLETE COMPUTING STACK → CPU AND MEMORY HIERARCHY → PROCESSOR AND EMBEDDED MAP → PROGRAM AND EXECUTION LADDER → NETWORK DEVICE AND LAYER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

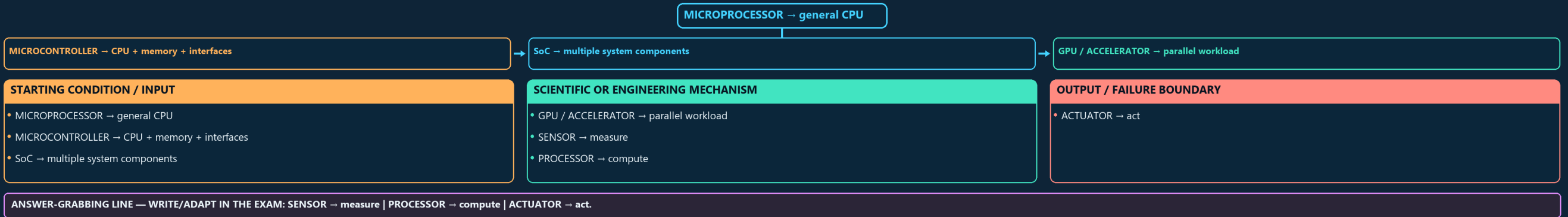
CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

STAGE 03 PROCESSOR AND EMBEDDED MAP

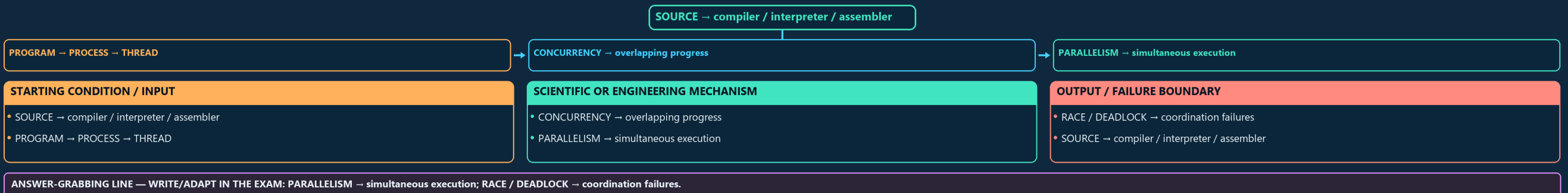
PROCESSOR AND EMBEDDED MAP GENERAL CPU CPU + MEMORY + INTERFACES MULTIPLE SYSTEM COMPONENTS PARALLEL WORKLOAD



04

STAGE 04 PROGRAM AND EXECUTION LADDER

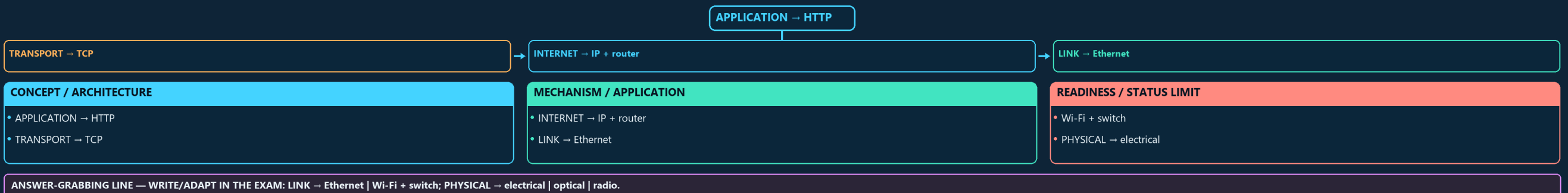
PROGRAM AND EXECUTION LADDER OVERLAPPING PROGRESS SIMULTANEOUS EXECUTION COORDINATION FAILURES



05

STAGE 05 NETWORK DEVICE AND LAYER MAP

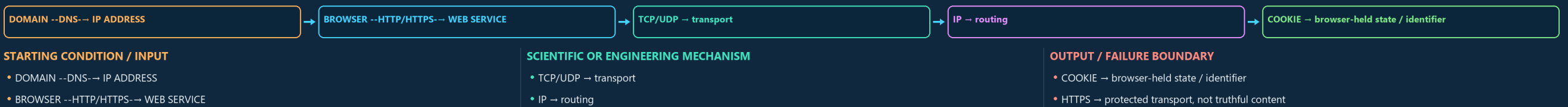
NETWORK DEVICE AND LAYER MAP IP + ROUTER WI-FI + SWITCH APPLICATION → HTTP



06

STAGE 06 INTERNET AND WEB REQUEST FLOW

INTERNET AND WEB REQUEST FLOW DOMAIN --DNS-- IP ADDRESS BROWSER --HTTP/HTTPS-- WEB SERVICE BROWSER-HELD STATE PROTECTED TRANSPORT, NOT TRUTHFUL CONTENT



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LINK → Ethernet | Wi-Fi + switch; PHYSICAL → electrical | optical | radio.

06

STAGE 06 INTERNET AND WEB REQUEST FLOW

INTERNET AND WEB REQUEST FLOW DOMAIN --DNS-- IP ADDRESS BROWSER --HTTP/HTTPS-- WEB SERVICE BROWSER-HELD STATE PROTECTED TRANSPORT, NOT TRUTHFUL CONTENT

DOMAIN --DNS-- IP ADDRESS

BROWSER --HTTP/HTTPS-- WEB SERVICE

TCP/UDP → transport

IP → routing

COOKIE → browser-held state / identifier

STARTING CONDITION / INPUT

- DOMAIN --DNS-- IP ADDRESS
- BROWSER --HTTP/HTTPS-- WEB SERVICE

SCIENTIFIC OR ENGINEERING MECHANISM

- TCP/UDP → transport
- IP → routing

OUTPUT / FAILURE BOUNDARY

- COOKIE → browser-held state / identifier
- HTTPS → protected transport, not truthful content

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COOKIE → browser-held state / identifier; HTTPS → protected transport, not truthful content.

07

STAGE 07 PERFORMANCE AND WIRELESS MATRIX

PERFORMANCE AND WIRELESS MATRIX ACHIEVED RATE DELAY VARIATION PACKET CELLULAR VOICE OVER LTE DISTINCT ROLES WIRELESS LABEL NO UNIVERSAL RANGE OR WALL PENETRATION

BANDWIDTH → capacity

LATENCY → delay

LTE → packet cellular

THROUGHPUT → achieved rate

JITTER → delay variation

VoLTE → voice over LTE

SYSTEM / DEVICE / INSTITUTION

- BANDWIDTH → capacity
- THROUGHPUT → achieved rate
- LATENCY → delay

DECISIVE TECHNICAL DISTINCTION

- JITTER → delay variation
- LTE → packet cellular
- VoLTE → voice over LTE

STATUS / UNIT / EVIDENCE LIMIT

- VLC → distinct roles
- WIRELESS LABEL → no universal range or wall penetration

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WIRELESS LABEL → no universal range or wall penetration.

08

STAGE 08 VIRTUALISATION AND CLOUD BOUNDARY

VIRTUALISATION AND CLOUD BOUNDARY VIRTUAL HARDWARE + GUEST OS SHARED HOST KERNEL MANAGES MANY SERVICES FIVE NIST CHARACTERISTICS RESPONSIBILITY SHIFTS

VM → virtual hardware + guest OS

CONTAINER → shared host kernel

ORCHESTRATION → manages many services

CLOUD → five NIST characteristics

SYSTEM / DEVICE / INSTITUTION

- VM → virtual hardware + guest OS
- CONTAINER → shared host kernel

DECISIVE TECHNICAL DISTINCTION

- ORCHESTRATION → manages many services
- CLOUD → five NIST characteristics

STATUS / UNIT / EVIDENCE LIMIT

- SaaS → responsibility shifts
- VM → virtual hardware + guest OS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: IaaS | PaaS | SaaS → responsibility shifts.

09

STAGE 09 DATA AND DISTRIBUTED TRADE-OFF

DATA AND DISTRIBUTED TRADE-OFF ORGANISED MANAGED DATA TRANSACTION PROPERTIES AVAILABILITY/LOCALITY, NOT BACKUP CAP DURING PARTITION CONSISTENCY OR AVAILABILITY TRADE PLACEMENT BY LATENCY AND RESILIENCE

SYSTEM / DEVICE / INSTITUTION

DATABASE / DBMS → organised managed data

ACID → transaction properties

DECISIVE TECHNICAL DISTINCTION

REPLICATION → availability/locality, not backup

CAP DURING PARTITION → consistency or availability trade

STATUS / UNIT / EVIDENCE LIMIT

EDGE / FOG / CLOUD → placement by latency and resilience

DATABASE / DBMS → organised managed data

SYSTEM / DEVICE / INSTITUTION

- DATABASE / DBMS → organised managed data
- ACID → transaction properties

DECISIVE TECHNICAL DISTINCTION

- REPLICATION → availability/locality, not backup
- CAP DURING PARTITION → consistency or availability trade

STATUS / UNIT / EVIDENCE LIMIT

- EDGE / FOG / CLOUD → placement by latency and resilience
- DATABASE / DBMS → organised managed data

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EDGE / FOG / CLOUD → placement by latency and resilience.

09

STAGE 09 DATA AND DISTRIBUTED TRADE-OFF

DATA AND DISTRIBUTED TRADE-OFF ORGANISED MANAGED DATA TRANSACTION PROPERTIES AVAILABILITY/LOCALITY, NOT BACKUP CAP DURING PARTITION CONSISTENCY OR AVAILABILITY TRADE PLACEMENT BY LATENCY AND RESILIENCE

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
DATABASE / DBMS → organised managed data	REPLICATION → availability/locality, not backup	EDGE / FOG / CLOUD → placement by latency and resilience
ACID → transaction properties	CAP DURING PARTITION → consistency or availability trade	DATABASE / DBMS → organised managed data
SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
• DATABASE / DBMS → organised managed data	• REPLICATION → availability/locality, not backup	• EDGE / FOG / CLOUD → placement by latency and resilience
• ACID → transaction properties	• CAP DURING PARTITION → consistency or availability trade	• DATABASE / DBMS → organised managed data

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EDGE / FOG / CLOUD → placement by latency and resilience.

10

STAGE 10 LEDGER CLAIM FIREWALL

LEDGER CLAIM FIREWALL DISTRIBUTED LEDGER REPLICATED RECORD LINKED BLOCKS + CONSENSUS ACCESS MODELS ORACLE INPUT CAN BE FALSE CONDITIONAL CLAIMS

BLOCKCHAIN → linked blocks + consensus	CONSORTIUM → access models	ORACLE INPUT → can be false
SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
• DISTRIBUTED LEDGER → replicated record	• CONSORTIUM → access models	• NFT / WEB3 / IMMUTABILITY → conditional claims
• BLOCKCHAIN → linked blocks + consensus	• ORACLE INPUT → can be false	• DISTRIBUTED LEDGER → replicated record

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NFT / WEB3 / IMMUTABILITY → conditional claims.

11

STAGE 11 ADVANCED COMPUTING AND STATUS RAIL

ADVANCED COMPUTING AND STATUS RAIL PROCESSOR + ACCELERATOR + INTERCONNECT + SOFTWARE SOFTWARE CAPABILITY DEVICE BASE QUBITS FOR SELECTED PROBLEMS CONFIDENTIALITY + INTEGRITY + AVAILABILITY VERIFIED OUTCOME

HPC → processor + accelerator + interconnect + software → AI → software capability → SEMICONDUCTOR → device base → QUANTUM → qubits for selected problems → SECURITY → confidentiality + integrity + availability				
CONCEPT / ARCHITECTURE		MECHANISM / APPLICATION		READINESS / STATUS LIMIT
• HPC → processor + accelerator + interconnect + software • AI → software capability		• SEMICONDUCTOR → device base • QUANTUM → qubits for selected problems		• SECURITY → confidentiality + integrity + availability • STANDARD → DEMO → DEPLOYMENT → VERIFIED OUTCOME

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STANDARD → DEMO → DEPLOYMENT → VERIFIED OUTCOME.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT MULTIPLE TASKS MAKE PROGRESS DURING OVERLAPPING TIME. TASKS EXECUTE SIMULTANEOUSLY. RACE CONDITION RESULT DEPENDS ON UNCONTROLLED EXECUTION ORDER. TASKS WAIT INDEFINITELY FOR RESOURCES HELD BY ONE ANOTHER. VIRTUAL MEMORY

MAPS PROCESS ADDRESSES TO PHYSICAL MEMORY/STORAGE, ENABLING ISOLATION AND

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- Concurrency: multiple tasks make progress during overlapping time.
- Parallelism: tasks execute simultaneously.
- Race condition: result depends on uncontrolled execution order.
- Deadlock: tasks wait indefinitely for resources held by one another.

READINESS, UNIT OR STATUS LIMIT

- Virtual memory: maps process addresses to physical memory/storage, enabling isolation and a
- Routing: selecting paths between networks.
- NAT: translates address information, commonly allowing many private devices to share a public
- DHCP: automatically supplies network configuration.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- Content Delivery Network: distributes cached content closer to users.
- Load balancer: spreads requests across servers.
- Zero trust: does not grant implicit trust merely because a user/device is inside a network;
- Cloud centralises compute and economies of scale.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.